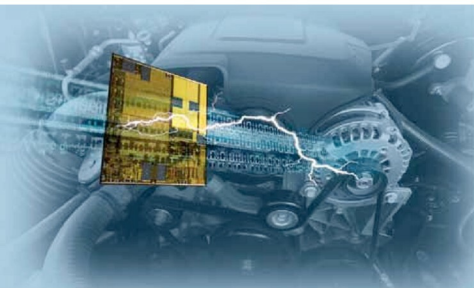


INDIA TURNING INTO R&D POWERHOUSE

More and more MNCs are flocking to India to set up an R&D base. Let's see what makes India a perfect destination for carrying out research activities from the standpoint of technology and innovation



■ **UMA GUPTA**

Today, India is looked at as one of the cheapest R&D destinations with highly talented engineers. The setting up of an R&D base by an increasing number of MNCs is a testimony to this fact. These R&D set-ups either serve the local market or help the parent company deliver new generation of products faster to the global market. For instance, one of the primary research areas for Philips Lighting India centre is to develop products that can suit Indian operating conditions.

In particular, embedded systems related work is growing fast and maturing to focus on enabling value-added devices. Almost every major international firm from Intel to Texas Instruments (TI) has its R&D or design lab for microprocessors in India. Telecom, consumer electronics, automation

and automotive are the top verticals with MNC R&D base in India.

"Automotive, healthcare and power are driving the demand in the semiconductor industry and we are well positioned to take advantage of the growing market by investing in R&D. The introduction of latest technologies will help us differentiate from our competitors," says Guruswamy Ganesh, vice president and country manager, Freescale Semiconductor India.

Infineon India contributes to almost every major automotive or smartcard product launched globally. The centre has aggressive plans for coming years. "As the market in India develops, the local R&D is expected to work closely with the marketing and sales teams to increase competitiveness of Infineon solutions for local markets. Existing engineering activities will be expanded further by initiating a new competence

centre for power electronics. Further, engineering of applications over Infineon products specific to Indian market is being considered," informs Vinay Shenoy, managing director, Infineon Technology India.

The immense intellectual power that India possesses, makes it a perfect destination for carrying out research activities from the standpoint of technology and innovation.

Samsung employs 5500 researchers in India and plans to hire 1100 more

MNC R&D in India: A Quick Look

1. Meant to serve either the global market or develop products suited for the Indian market
2. Operating cost in India 25 per cent lower than in China
3. Tier-II locations like Chandigarh and Bhubaneswar emerging fast by offering up to 40-50 per cent savings on cost
4. Embedded systems related work growing fast
5. Automotive, healthcare, power and security driving the demand in semiconductors
6. Local advantages: Availability of highly skilled professionals in hardware and software, lower operating cost and growing Indian market
7. Local deterrents: Talent crunch, lack of adequate physical infrastructure, not-so-encouraging industry-academia interface
8. Suggestions:
 - Engineering institutes in India should impart postgraduate education in technologies relevant for the India market, like energy efficiency, automotive safety and security
 - The government should provide encouraging tax incentives to help companies grow their R&D labs here